

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

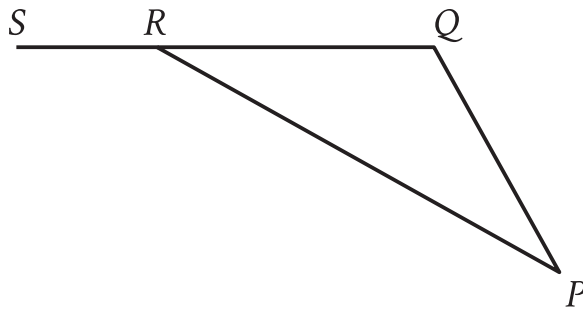
Lines, angles, and triangles

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

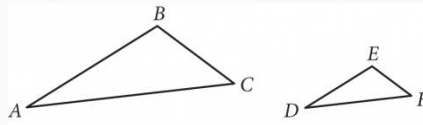
- Line segment QR is extended to point S .
- A note indicates the figure is not drawn to scale.

In triangle PQR , QR is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

- A. 48°
- B. 31°
- C. 24°
- D. 17°

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12



Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20° , what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

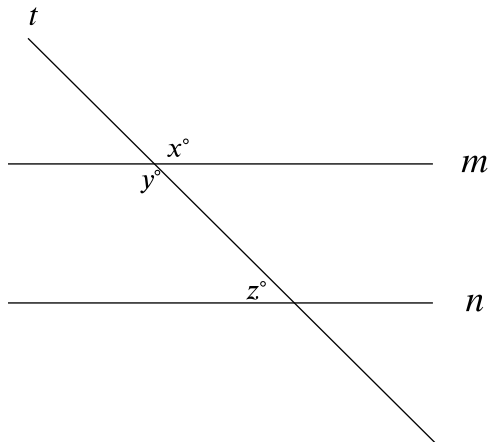
STUDENT-PRODUCED RESPONSE

○	/○	/○	○
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.



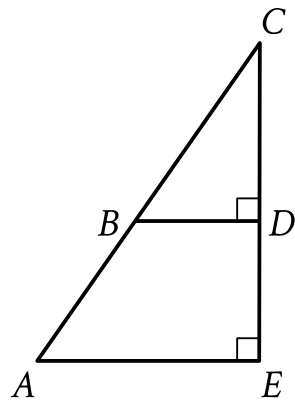
Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 2 angles are labeled clockwise from top left as follows:
 - Top right: x°
 - Bottom left: y°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Top left: z°
- A note indicates the figure is not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

- A. 3
- B. 21
- C. 41

D. 139



Note: Figure not drawn to scale.

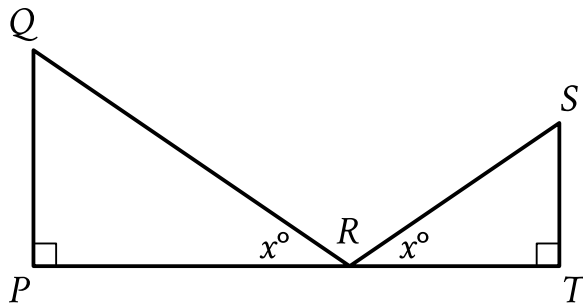
- Line segment BD is drawn from side CA to side CE to form triangle CBD .
- Angle CDB is a right angle.
- Angle CEA is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26BD$. What is the measure of angle CAE ?

- A. $26 \cdot 57^\circ$
- B. $26 + 57^\circ$
- C. 57°
- D. 26°

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°



Note: Figure not drawn to scale.

- The 2 triangles share common vertex upper R.
 - Common vertex upper R lies on line segment upper P upper T.
- Triangle upper Q upper P upper R is labeled as follows:
 - Angle upper Q upper P upper R is a right angle.
 - The measure of angle upper Q upper R upper P is x° .
- Triangle upper S upper T upper R is labeled as follows:
 - Angle upper S upper T upper R is a right angle.
 - The measure of angle upper S upper R upper T is x° .
- A note indicates the figure is not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\bar{ST}$, $\bar{QP}$, $\bar{PR}$, and $\bar{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\bar{SR}$?

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

Q9

GRID-IN

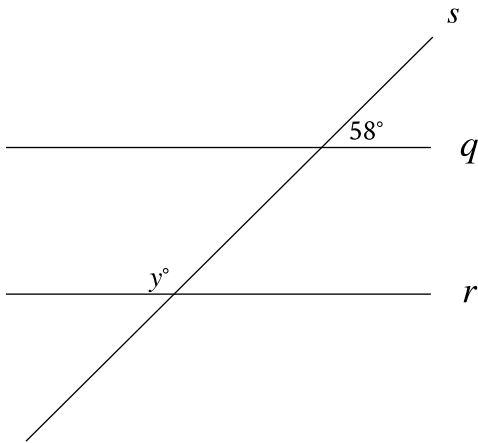
GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled s , q , and r .
- Line s intersects both line q and line r .
- At the intersection of line s and line q , 1 angle is labeled clockwise from top left as follows:
 - Top right: 58°
- At the intersection of line s and line r , 1 angle is labeled clockwise from top left as follows:
 - Top left: y°
- A note indicates the figure is not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.